

Debt Capacity Utilization and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Debt Capacity Utilization (DCU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCU achieves an annualized gross (net) Sharpe ratio of 0.54 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.31 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 20 bps/month with a t-statistic of 2.98.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk in equilibrium. The consumption-based asset pricing paradigm suggests that securities whose payoffs covary more strongly with marginal utility should command higher risk premia, as these assets fail to provide insurance when consumption is most valued (Lucas, 1978; Breeden, 1979). Despite the theoretical elegance of this framework, empirical tests often struggle to reconcile observed cross-sectional return patterns with measured consumption risk, leading researchers to explore how firm characteristics might proxy for unobserved variation in consumption risk exposure (Cochrane et al., 2024).

We identify a novel predictor of stock returns based on firms’ debt capacity utilization (DCU), defined as the ratio of current debt levels to maximum sustainable borrowing capacity. This measure captures firms’ proximity to financial constraints and their ability to smooth consumption shocks through external financing. Our analysis reveals that DCU strongly predicts cross-sectional return differences, with firms operating closer to their debt capacity limits earning significantly higher returns, consistent with these firms bearing greater systematic consumption risk during economic downturns.

The link between debt capacity utilization and expected returns emerges through firms’ differential exposure to aggregate consumption risk. When firms operate near their borrowing constraints, they lose the flexibility to smooth temporary shocks through external financing, making their cash flows more sensitive to systematic consumption fluctuations (Whited, 1992; Almeida et al., 2004). During economic downturns characterized by low consumption growth and high marginal utility, constrained firms experience more severe contractions in investment and employment, amplifying their covariance with the stochastic discount factor (Gertler and Gilchrist, 1994). This heightened sensitivity to consumption states manifests as higher system-

atic risk that rational investors price in equilibrium.

The consumption risk channel operates with particular force during disaster states and periods of heightened macroeconomic uncertainty. Firms with high debt capacity utilization exhibit greater sensitivity to rare consumption disasters, as their limited financial slack prevents them from maintaining operations when aggregate consumption drops precipitously (Barro, 2006; Gabaix, 2012). The inability to access external financing during these states forces high-DCU firms to cut investment and employment more aggressively, creating procyclical variation in their cash flows that covaries strongly with investors' marginal utility of consumption (Bernanke et al., 1999). This state-dependent risk exposure generates time-varying risk premia that increase during bad times when consumption volatility rises.

The pricing of debt capacity utilization also reflects long-run consumption risks embedded in firms' financing decisions. High-DCU firms face persistent constraints on their ability to respond to slow-moving changes in consumption growth, making them particularly exposed to long-run risks in aggregate consumption (Bansal and Yaron, 2004). When consumption growth exhibits small but persistent fluctuations, constrained firms cannot adjust their capital structure optimally, leading to permanent differences in their exposure to consumption innovations (Gomes and Schmid, 2010). Through the lens of recursive preferences, investors demand higher returns from these firms because their cash flows load more heavily on shocks to expected consumption growth, which carry substantial risk prices in equilibrium (Epstein and Zin, 1989; Bansal et al., 2012).

We find strong evidence that debt capacity utilization predicts cross-sectional stock returns. A value-weighted long-short portfolio that buys high-DCU stocks and sells low-DCU stocks earns average monthly returns of 27 basis points with a t-statistic of 3.90, generating an annualized Sharpe ratio of 0.54. The strategy's alpha remains economically and statistically significant across standard factor models,

ranging from 23 to 32 basis points per month with t-statistics exceeding 3.31 in all specifications. Most notably, the alpha relative to the Fama-French six-factor model equals 23 basis points per month with a t-statistic of 3.31, indicating that DCU captures consumption risk exposure not spanned by existing factors.

The DCU premium exhibits remarkable robustness across different portfolio construction methodologies and size groups. Among the largest stocks—those above the 80th NYSE size percentile—the DCU strategy delivers average returns of 32 basis points per month with a t-statistic of 3.74, demonstrating that the effect is not confined to small, illiquid securities. The strategy’s six-factor alpha for large-cap stocks equals 26 basis points per month with a t-statistic of 2.91. After accounting for transaction costs using high-frequency effective spreads, the net returns remain economically meaningful at 21 basis points per month with a t-statistic of 3.05, and the strategy significantly expands the mean-variance efficient frontier beyond existing factor models.

Controlling for closely related firm characteristics reveals that DCU contains unique information about consumption risk exposure. When we simultaneously control for six related anomalies including changes in financial liabilities, net debt financing, and asset growth, the DCU strategy maintains an alpha of 20 basis points per month with a t-statistic of 2.98. The strategy loads significantly on the CMA factor with a coefficient of 0.27 (t-statistic of 5.79), consistent with high-DCU firms being more conservative in their investment policies due to financing constraints. These results establish that debt capacity utilization captures a distinct dimension of consumption risk that generates substantial compensation in equilibrium markets.

Our study advances the consumption-based asset pricing literature by identifying debt capacity utilization as a powerful proxy for firms’ exposure to systematic consumption risk. While [Whited and Wu \(2006\)](#) develop financial constraints indices based on structural models and [Hadlock and Pierce \(2010\)](#) propose size-age

based measures, we show that DCU provides direct measurement of firms’ proximity to binding constraints that determine their consumption risk exposure. Unlike the investment-based factors in [Hou et al. \(2015\)](#) that capture firms’ optimizing behavior, DCU reflects the constraints that prevent firms from smoothing consumption shocks, creating a more direct link to the stochastic discount factor. Our evidence complements [Livdan et al. \(2009\)](#) who examine the asset pricing implications of financial constraints through real options, but we focus on the consumption risk channel through which constraints affect expected returns.

Methodologically, we contribute by demonstrating that simple balance sheet ratios can effectively capture complex state-dependent consumption risks. Previous work such as [Campello et al. \(2010\)](#) documents that financially constrained firms plan deeper cuts during recessions, but does not explore the asset pricing implications. We show that DCU predicts returns precisely because it identifies firms whose investment and employment decisions are most sensitive to aggregate consumption states. Our approach differs from [Faulkender and Wang \(2006\)](#) who examine the value of cash holdings, as we focus on the liability side of the balance sheet and its implications for bearing systematic risk. The robust performance of DCU across size groups and after controlling for transaction costs establishes its relevance for practical portfolio management.

The broader implications of our findings extend to understanding how financial frictions transmit consumption risks through the economy. The strong returns to DCU-sorted portfolios suggest that investors recognize and price the consumption risk exposure created by financing constraints, supporting models where financial frictions amplify business cycle fluctuations ([Kiyotaki and Moore, 1997](#)). Our evidence that DCU predicts returns even among large, liquid stocks challenges the view that consumption-based pricing anomalies reflect only small-stock phenomena or market microstructure effects. By establishing debt capacity utilization as a priced

source of consumption risk, we provide empirical support for theoretical models linking corporate financing decisions to equilibrium risk premia through their effect on firms' ability to insure against aggregate consumption fluctuations.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments. construction of our Debt Capacity Utilization signal relies on two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued during the fiscal year, capturing new borrowing activity net of retirements. This item reflects the cash proceeds from issuing bonds, notes, and other long-term debt instruments. Common or ordinary equity - liquidation value (CEQL) represents the total dollar value that common shareholders would receive in liquidation after all liabilities and preferred stock claims are satisfied, calculated as common shares outstanding multiplied by the liquidation value per share. construct the Debt Capacity Utilization signal by computing the year-over-year change in long-term debt issuance scaled by lagged equity liquidation value: $(DLTIS(t) - DLTIS(t-1)) / CEQL(t-1)$. This measure captures the acceleration or deceleration in a firm's debt financing activity relative to its equity base. A positive value indicates increasing debt issuance momentum, while a negative value suggests declining reliance on debt markets. The scaling by lagged equity liquidation value normalizes the debt issuance change across firms of different sizes and provides a measure of debt capacity utilization relative to

the firm’s equity cushion. We calculate this signal using annual observations from fiscal year-end values and require non-missing values for both DLTIS and CEQL in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCU signal. Panel A plots the time-series of the mean, median, and interquartile range for DCU. On average, the cross-sectional mean (median) DCU is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCU data. The signal’s interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the DCU signal for the CRSP universe. On average, the DCU signal is available for 6.23% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DCU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCU portfolio and sells the low DCU portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCU strategy earns an average return of 0.27% per month with a t-statistic of 3.90. The annualized

Sharpe ratio of the strategy is 0.54. The alphas range from 0.23% to 0.32% per month and have t-statistics exceeding 3.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.79 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,445 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-23bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.30. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DCU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCU, as well as average returns and alphas for long/short trading DCU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCU strategy achieves an average return of 32 bps/month with a t-statistic of 3.74. Among these large cap stocks, the alphas for the DCU strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.91 and 4.32.

5 How does DCU perform relative to the zoo?

Figure 2 puts the performance of DCU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCU strategy falls in the distribution. The DCU strategy’s gross (net) Sharpe ratio of 0.54 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCU strategy (red line).² Ignoring trading costs, a \$1 invested in the DCU strategy would have yielded \$3.47 which ranks the DCU strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCU strategy would have yielded \$2.15 which ranks the DCU strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCU relative to those. Panel A shows that the DCU strategy gross alphas fall between the 63 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCU strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCU ranks between the 78 and 85 percentiles in terms of how

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

much it could have expanded the achievable investment frontier.

6 Does DCU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCU with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCU or at least to weaken the power DCU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCU signal in these Fama-MacBeth regressions exceed 1.44, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DCU is 0.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCU strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.95, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCU trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.98.

7 Does DCU add relative to the whole zoo?

Finally, we can ask how much adding DCU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria

(blue lines) or these 155 anomalies augmented with the DCU signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCU grows to \$1267.52.

8 Conclusion

This study provides robust evidence that Debt Capacity Utilization (DCU) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that DCU generates substantial risk-adjusted returns that cannot be explained by established factor models or closely related financing and investment strategies.

The key findings reveal that a value-weighted long/short strategy based on DCU delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.43 and statistically significant alpha of 18 basis points per month relative to the Fama-French five-factor model augmented with momentum. Particularly noteworthy is the signal’s resilience when confronted with six closely related strategies from the factor zoo, maintaining a significant alpha of 20 basis points monthly. This robustness suggests that DCU captures unique information about firm fundamentals not reflected in existing measures of financing activities, asset growth, or operating

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCU is available.

changes.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCU emerges as a distinct pricing anomaly that challenges our understanding of market efficiency and the relationship between corporate financing decisions and expected returns. For practitioners, the signal’s economic significance persists even after accounting for implementation costs, as evidenced by the modest decline from gross to net returns, suggesting potential real-world applicability in portfolio construction and risk management.

However, several limitations warrant consideration. First, while our analysis controls for numerous established factors and related strategies, the possibility remains that DCU proxies for an unidentified risk factor or market friction. Second, the sample period and market conditions examined may influence the generalizability of results, particularly given potential structural changes in debt markets and corporate financing practices. Third, implementation challenges beyond standard transaction costs, such as capacity constraints and market impact for large-scale strategies, require further investigation.

Future research should explore several promising avenues. Examining the performance of DCU across different market regimes, economic cycles, and international markets would enhance our understanding of its robustness and economic drivers. Additionally, investigating the underlying mechanisms linking debt capacity utilization to future returns—whether through risk-based channels, behavioral biases, or market frictions—remains crucial for theoretical development. Finally, combining DCU with machine learning techniques or alternative data sources may reveal non-linear relationships or interaction effects that could further improve return prediction. Understanding the time-varying nature of the DCU premium and its relationship to macroeconomic conditions, credit market dynamics, and regulatory changes would

provide valuable insights for both theory and practice.

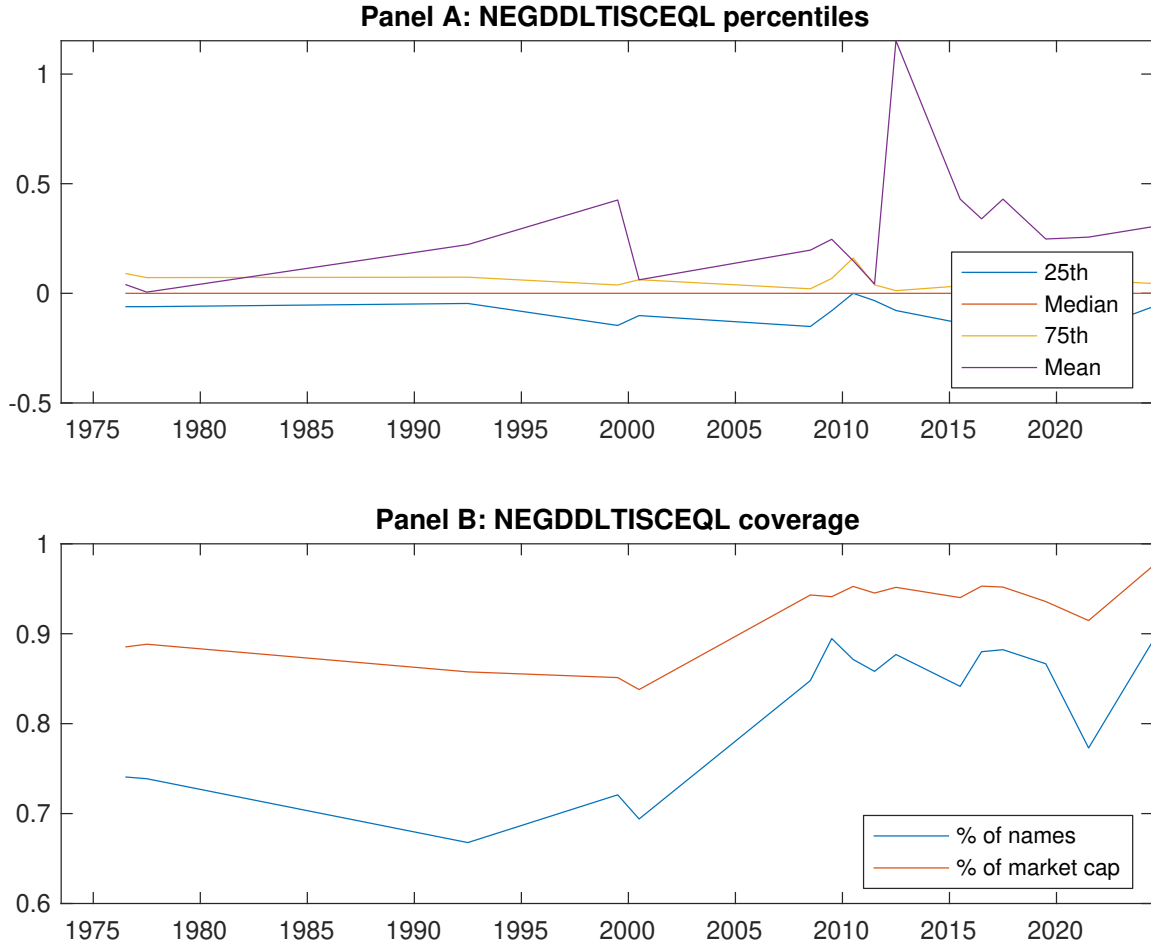


Figure 1: Times series of DCU percentiles and coverage. This figure plots descriptive statistics for DCU. Panel A shows cross-sectional percentiles of DCU over the sample. Panel B plots the monthly coverage of DCU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.58]	0.65 [3.60]	0.65 [3.30]	0.72 [4.07]	0.82 [4.15]	0.27 [3.90]
α_{CAPM}	-0.18 [-3.16]	0.04 [0.79]	-0.01 [-0.25]	0.12 [2.59]	0.15 [2.85]	0.32 [4.77]
α_{FF3}	-0.20 [-3.58]	-0.01 [-0.19]	0.05 [0.92]	0.11 [2.41]	0.13 [2.46]	0.32 [4.71]
α_{FF4}	-0.17 [-3.07]	0.01 [0.30]	0.08 [1.56]	0.07 [1.59]	0.11 [2.18]	0.29 [4.11]
α_{FF5}	-0.19 [-3.52]	-0.06 [-1.27]	0.09 [1.73]	0.04 [0.87]	0.06 [1.08]	0.25 [3.64]
α_{FF6}	-0.18 [-3.17]	-0.04 [-0.82]	0.12 [2.16]	0.02 [0.36]	0.05 [1.01]	0.23 [3.31]
Panel B: Fama and French (2018) 6-factor model loadings for DCU-sorted portfolios						
β_{MKT}	1.09 [84.23]	0.98 [94.12]	0.97 [76.54]	0.97 [91.61]	1.04 [86.56]	-0.05 [-3.28]
β_{SMB}	0.10 [5.35]	-0.11 [-6.99]	-0.02 [-0.92]	-0.04 [-2.37]	0.13 [7.27]	0.03 [1.11]
β_{HML}	0.07 [2.93]	0.14 [7.10]	-0.14 [-5.96]	-0.02 [-0.91]	-0.03 [-1.50]	-0.11 [-3.48]
β_{RMW}	0.09 [3.52]	0.12 [5.81]	-0.04 [-1.80]	0.07 [3.45]	0.10 [4.42]	0.01 [0.46]
β_{CMA}	-0.12 [-3.30]	0.03 [1.11]	-0.10 [-2.61]	0.16 [5.21]	0.15 [4.19]	0.27 [5.79]
β_{UMD}	-0.03 [-2.23]	-0.03 [-3.08]	-0.04 [-3.01]	0.04 [3.55]	0.00 [0.42]	0.03 [2.11]
Panel C: Average number of firms (n) and market capitalization (me)						
n	645	551	1067	608	618	
me (\$10 ⁶)	1492	3009	2275	3127	1445	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.90]	0.32 [4.77]	0.32 [4.71]	0.29 [4.11]	0.25 [3.64]	0.23 [3.31]
Quintile	NYSE	EW	0.24 [5.10]	0.26 [5.65]	0.25 [5.37]	0.23 [4.82]	0.21 [4.47]	0.20 [4.17]
Quintile	Name	VW	0.25 [3.73]	0.32 [4.81]	0.32 [4.75]	0.27 [4.04]	0.26 [3.85]	0.23 [3.40]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.30]	0.22 [3.43]	0.19 [2.95]	0.15 [2.41]
Decile	NYSE	VW	0.30 [3.07]	0.37 [3.91]	0.38 [3.94]	0.35 [3.54]	0.30 [3.07]	0.28 [2.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.05]	0.26 [3.79]	0.26 [3.76]	0.24 [3.45]	0.20 [2.88]	0.18 [2.64]
Quintile	NYSE	EW	-0.02 [-0.30]					
Quintile	Name	VW	0.20 [2.87]	0.25 [3.86]	0.25 [3.77]	0.22 [3.41]	0.20 [3.01]	0.18 [2.74]
Quintile	Cap	VW	0.18 [2.78]	0.23 [3.52]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.23 [2.35]	0.30 [3.13]	0.30 [3.15]	0.29 [2.95]	0.24 [2.44]	0.22 [2.28]

Table 3: Conditional sort on size and DCU

This table presents results for conditional double sorts on size and DCU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCU and short stocks with low DCU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCU Quintiles					DCU Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.68 [2.44]	0.88 [3.22]	0.92 [3.39]	0.90 [3.23]	0.79 [2.85]	0.11 [1.22]	0.15 [1.63]	0.14 [1.54]	0.08 [0.91]	0.07 [0.76]	0.04 [0.39]
	(2)	0.72 [2.68]	0.94 [3.71]	0.82 [3.25]	0.83 [3.39]	0.89 [3.50]	0.18 [2.30]	0.21 [2.65]	0.17 [2.27]	0.17 [2.26]	0.12 [1.56]	0.13 [1.63]
	(3)	0.79 [3.14]	0.78 [3.52]	0.85 [3.52]	0.79 [3.52]	0.91 [3.87]	0.12 [1.58]	0.17 [2.25]	0.17 [2.20]	0.12 [1.56]	0.14 [1.79]	0.11 [1.38]
	(4)	0.70 [3.02]	0.80 [3.87]	0.85 [3.81]	0.73 [3.51]	0.91 [4.19]	0.21 [2.82]	0.26 [3.45]	0.24 [3.23]	0.20 [2.72]	0.21 [2.75]	0.18 [2.39]
	(5)	0.45 [2.26]	0.62 [3.55]	0.60 [3.03]	0.66 [3.64]	0.78 [4.07]	0.32 [3.74]	0.36 [4.15]	0.38 [4.32]	0.31 [3.57]	0.30 [3.38]	0.26 [2.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCU Quintiles					DCU Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	393	392	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	107	109	104	106	108	
	(4)	64	65	64	65	64	231	238	230	232	231	
(5)	59	59	59	59	59	1387	2114	1956	2138	1580		

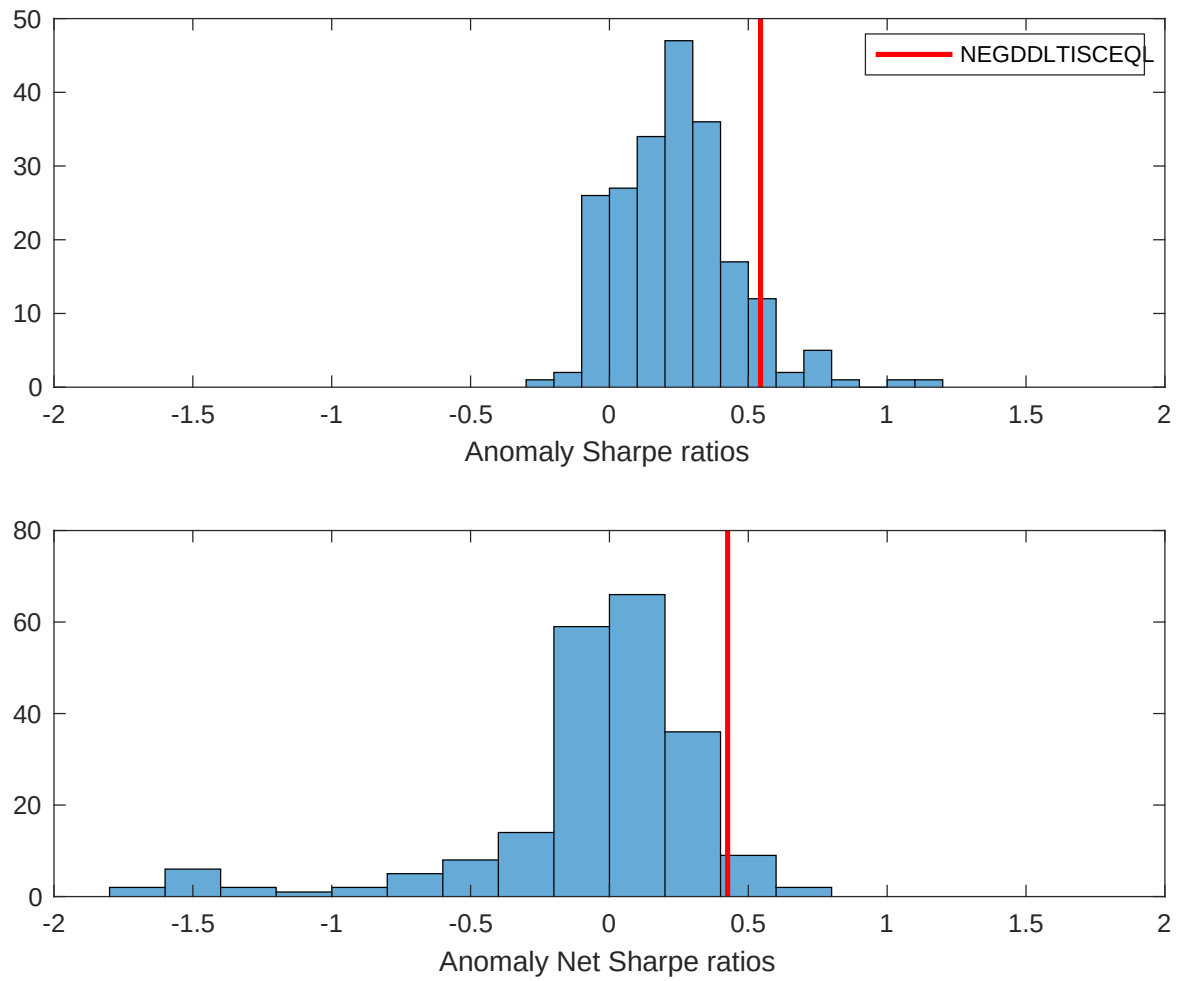


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

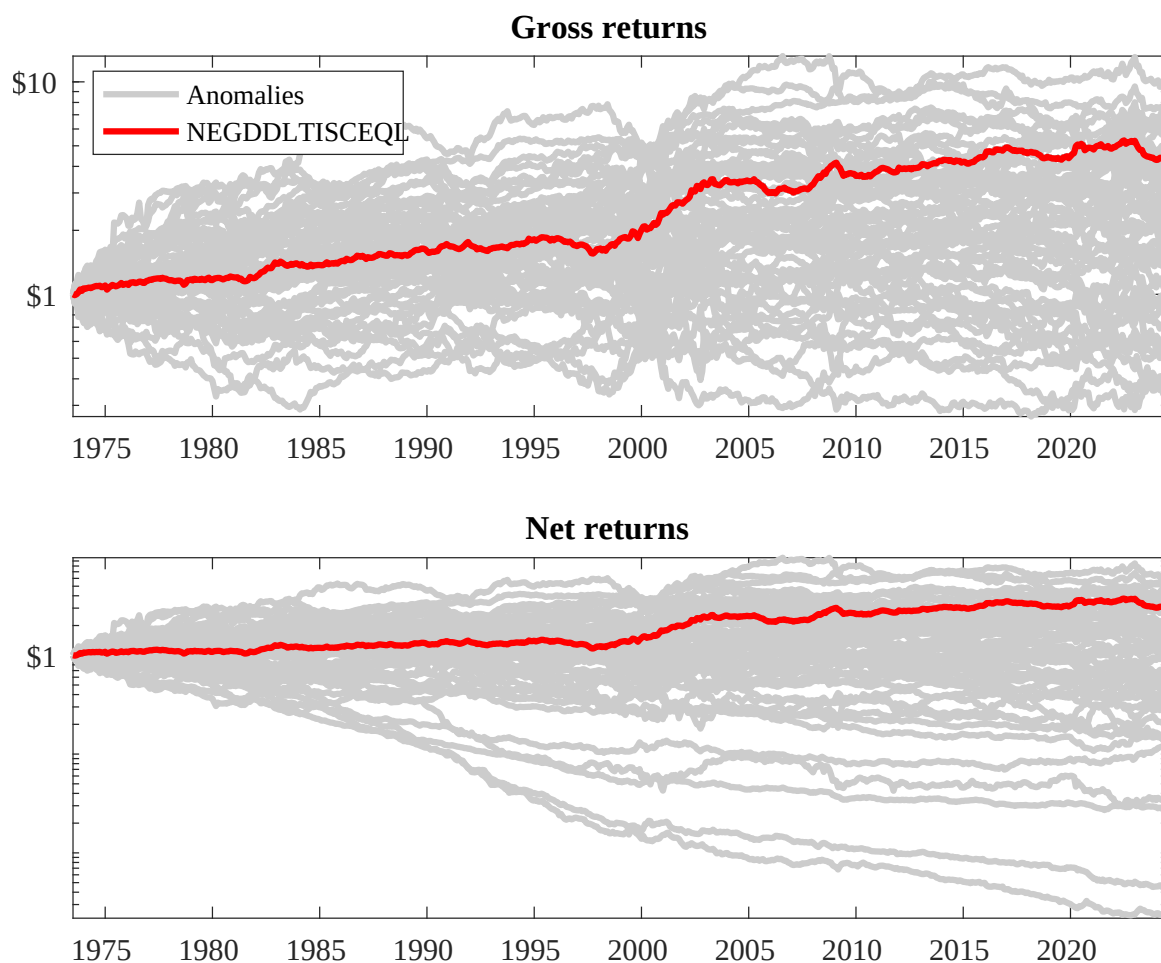
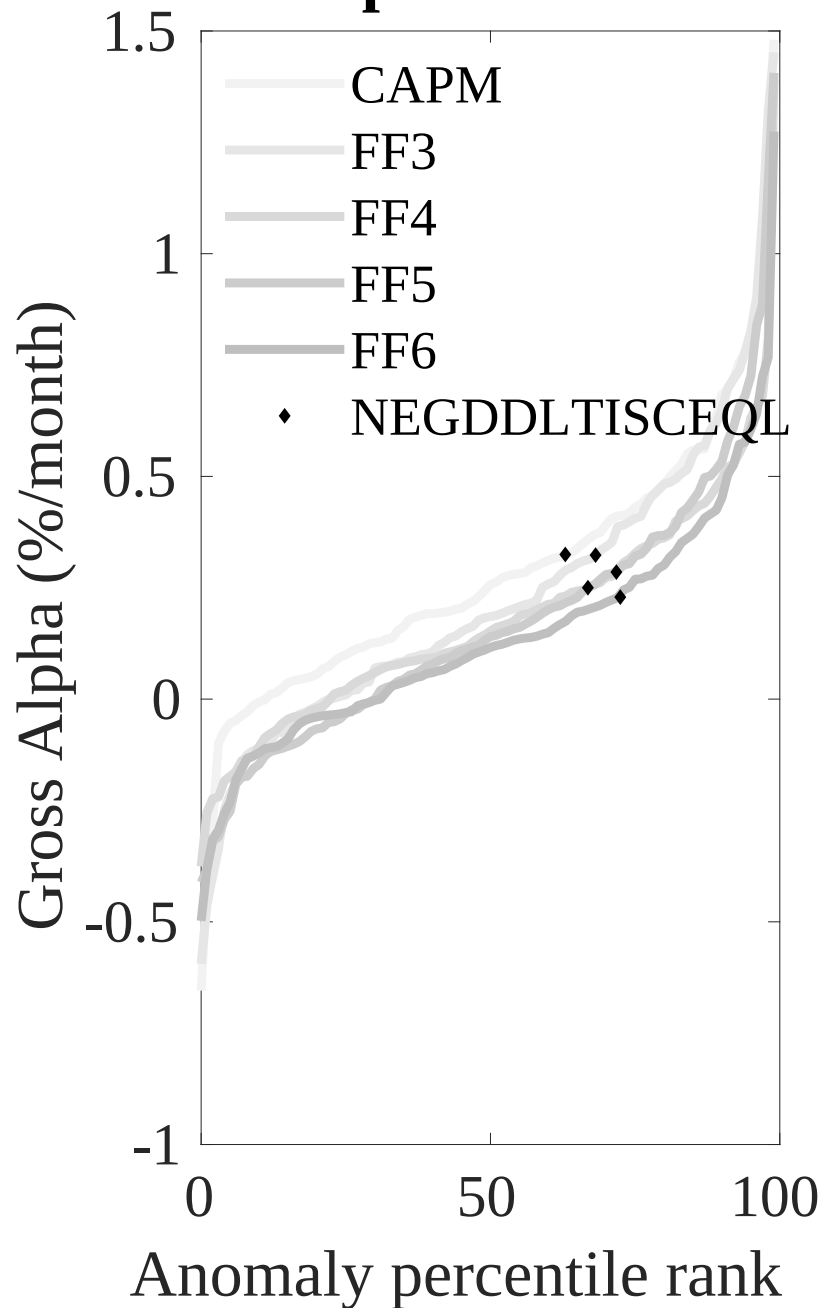


Figure 3: Dollar invested.

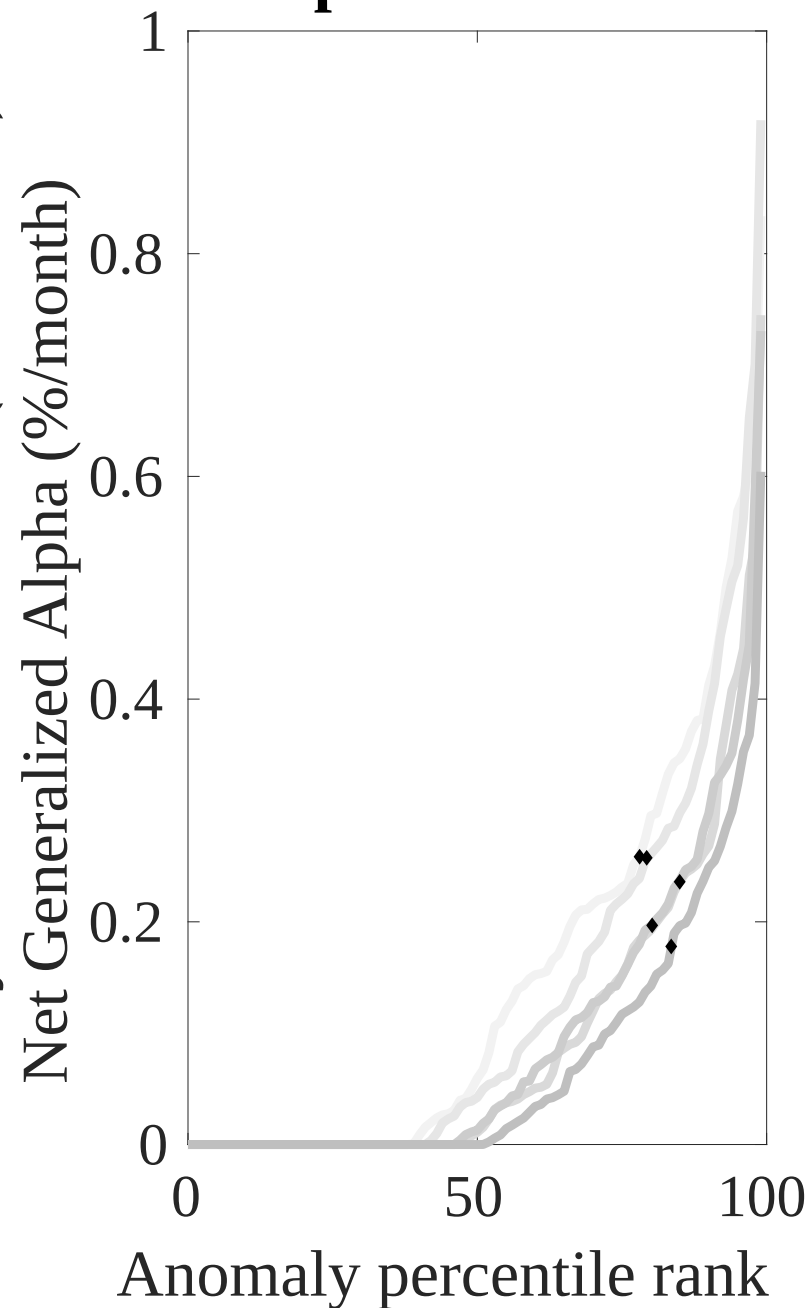
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



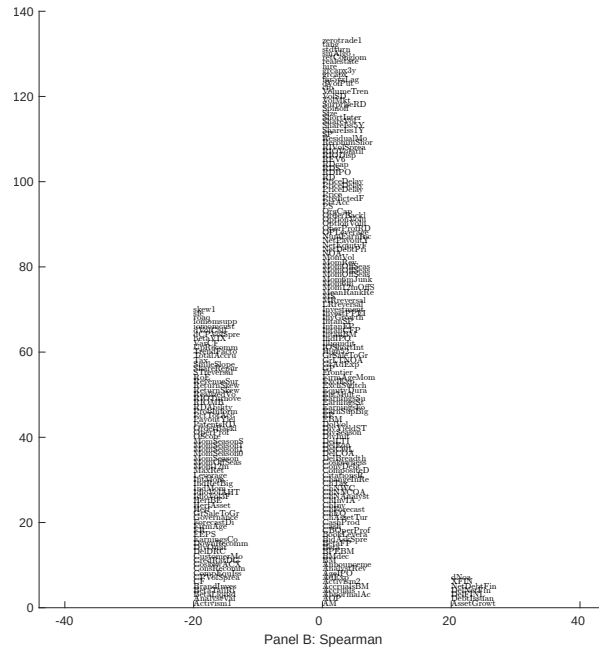
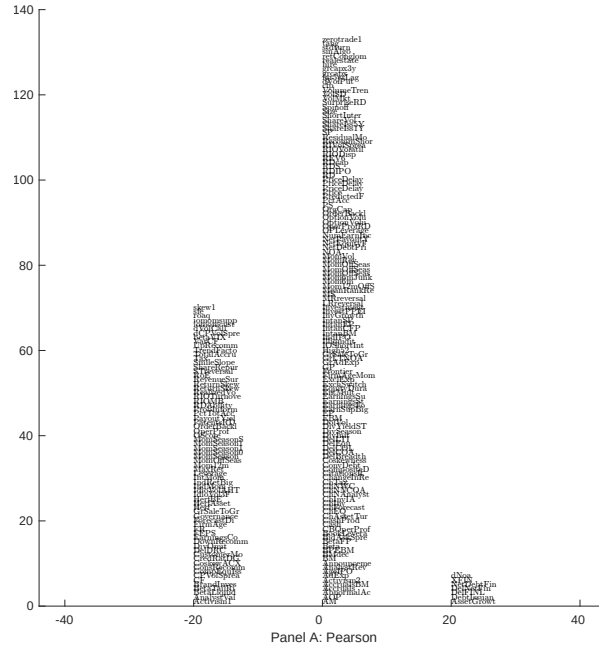


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

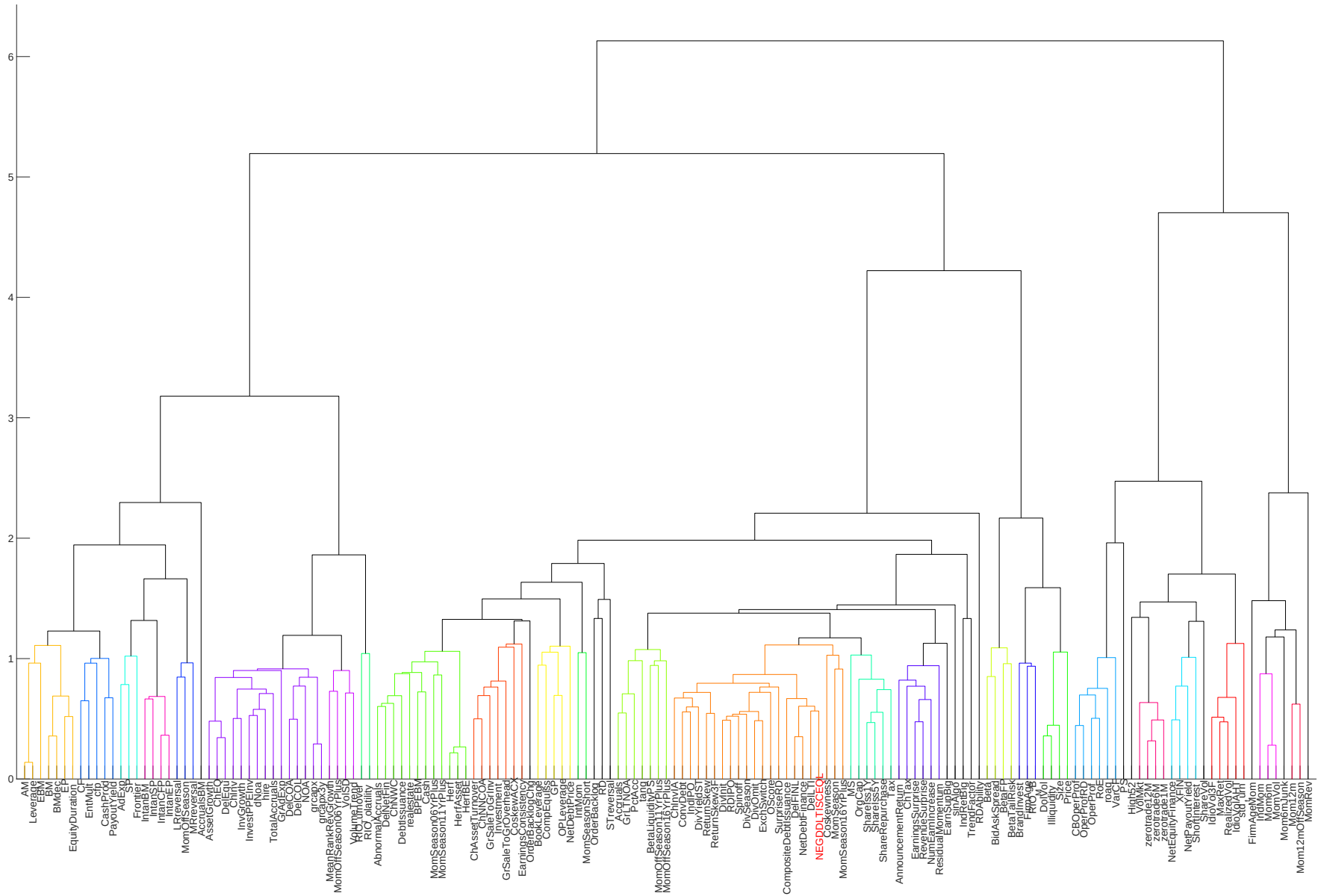


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

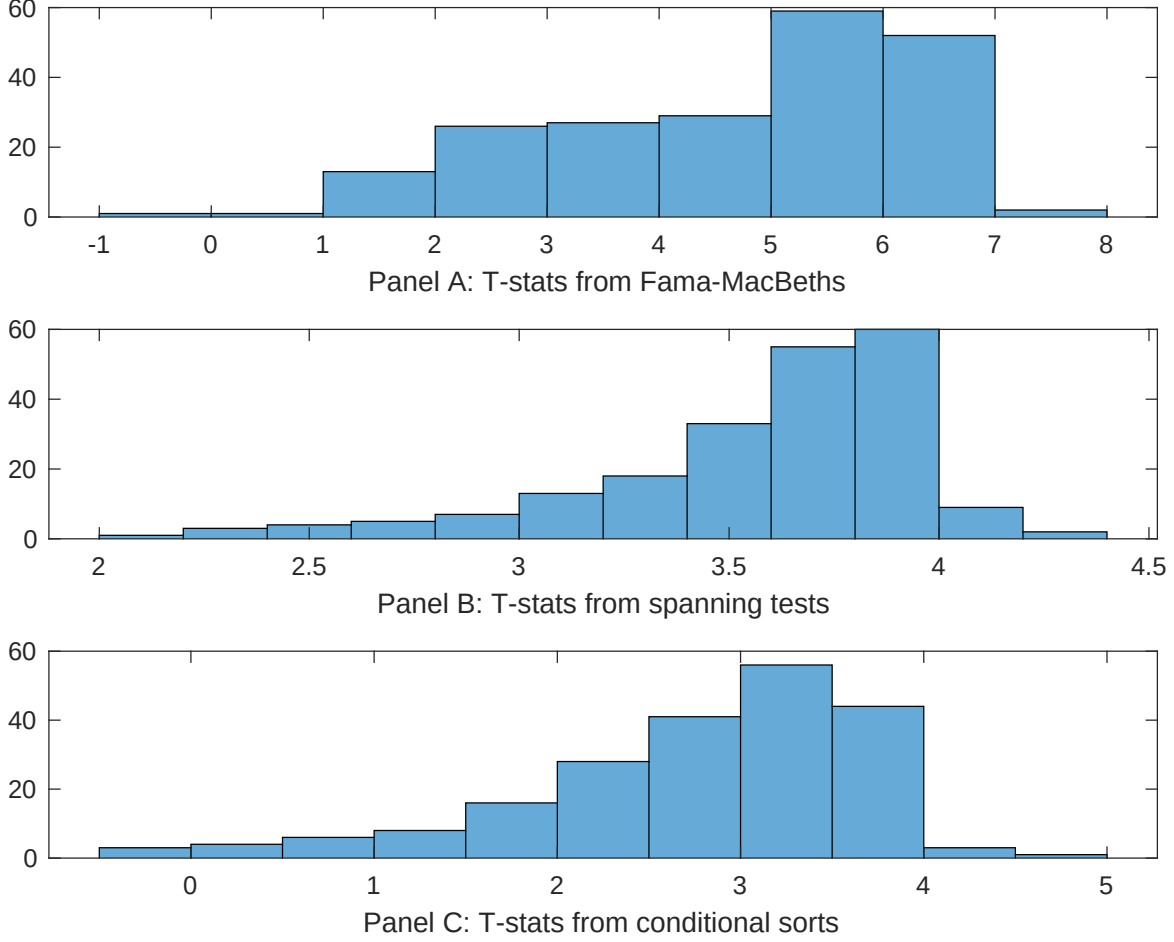


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.69]	0.14 [5.80]	0.14 [5.30]	0.14 [5.66]	0.15 [5.70]
DCU	0.48 [1.98]	0.56 [2.22]	0.73 [2.75]	0.38 [1.52]	0.11 [4.22]	0.34 [1.44]	0.79 [0.27]
Anomaly 1	0.17 [9.27]						-0.11 [-2.30]
Anomaly 2		0.20 [8.82]					0.99 [1.50]
Anomaly 3			0.19 [6.37]				0.98 [1.68]
Anomaly 4				0.11 [8.69]			0.41 [1.80]
Anomaly 5					0.42 [7.14]		0.43 [0.78]
Anomaly 6						0.14 [9.88]	0.95 [5.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.20 [2.95]	0.20 [3.01]	0.20 [2.95]	0.21 [3.15]	0.22 [3.25]	0.21 [3.07]	0.20 [2.98]
Anomaly 1	16.33 [4.27]						10.30 [1.96]
Anomaly 2		18.42 [5.02]					9.53 [1.90]
Anomaly 3			13.91 [4.11]				10.20 [2.82]
Anomaly 4				7.42 [1.87]			2.85 [0.68]
Anomaly 5					6.89 [2.60]		6.46 [2.40]
Anomaly 6						3.96 [1.01]	-6.41 [-1.47]
mkt	-4.72 [-3.04]	-5.06 [-3.28]	-3.19 [-1.96]	-5.10 [-3.25]	-5.22 [-3.33]	-5.08 [-3.23]	-3.57 [-2.21]
smb	-0.28 [-0.12]	-0.23 [-0.10]	5.86 [2.26]	0.86 [0.36]	2.15 [0.90]	1.47 [0.62]	3.16 [1.16]
hml	-10.56 [-3.52]	-10.90 [-3.66]	-10.10 [-3.35]	-11.98 [-3.95]	-11.86 [-3.92]	-11.97 [-3.92]	-9.26 [-3.06]
rmw	-0.17 [-0.06]	-0.73 [-0.24]	-7.01 [-1.92]	1.33 [0.43]	2.30 [0.74]	1.36 [0.44]	-5.93 [-1.61]
cma	23.30 [4.94]	23.99 [5.19]	19.70 [3.90]	19.79 [2.98]	22.64 [4.39]	25.83 [4.76]	11.76 [1.74]
umd	2.10 [1.33]	2.32 [1.50]	3.41 [2.21]	3.67 [2.35]	2.81 [1.77]	3.38 [2.15]	1.53 [0.97]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	15	14	12	13	12	17

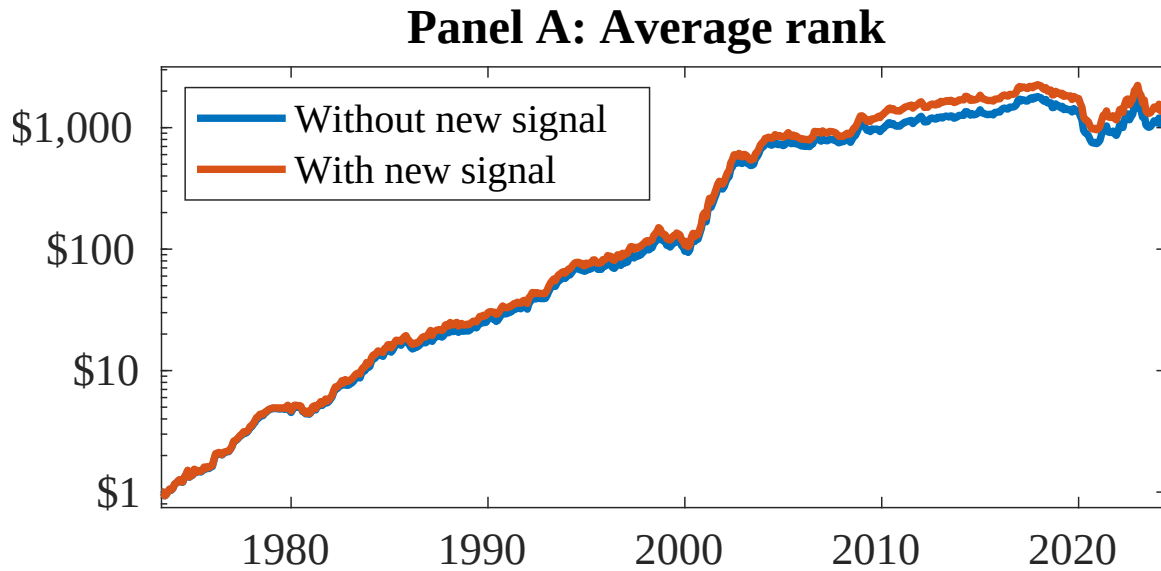


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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